

ALL NINE PROGRESSIONS OF THE CONJECTURE OF DAS, MAITY AND SAIKIA ON GENERALIZED CUBIC PARTITIONS MODULO PRIME SQUARES PRINTED IN ARXIV:2503.19399V2 FAIL

ABSTRACT. Let $a_c(n)$ be defined by $\sum_{n \geq 0} a_c(n) q^n = 1/(f_1 f_2^{c-1})$, where $f_k = \prod_{m \geq 1} (1 - q^{km})$. In the concluding remarks of the e-print arXiv:2503.19399v2 of their paper on generalized cubic and overcubic partitions, Das, Maity and Saikia conjecture nine arithmetic progressions of the shape $a_{p-6}(p^2 n + r) \equiv 0 \pmod{p^2}$, for $p \in \{31, 43, 59, 71, 83\}$. Every one of the nine is false as printed there. The cheapest witness is

$$a_{65}(41) = 451303321502143296879 \equiv 2170 \not\equiv 0 \pmod{71^2},$$

which kills the fourth line at $n = 0$, the smallest index its own quantifier admits; the other eight lines hold at $n = 0$ and fail at $n = 1$. The witness needs no computer: modulo 71 the coefficient equals the 41st coefficient of f_2^7/f_1 , which is a 21-term dot product of two published integer sequences and equals -31 , and $71 \nmid 31$. The refuted text is the e-print; the paywalled version of record has not been read.

1. THE STATEMENT

For an integer $c \geq 1$ put

$$(1) \quad \sum_{n \geq 0} a_c(n) q^n = \frac{1}{f_1 f_2^{c-1}}, \quad f_k := \prod_{m \geq 1} (1 - q^{km}), \quad a_c(0) = 1,$$

which is the displayed generating function of [2]: one uncoloured copy of every part, together with $c - 1$ further colours for the *even* parts only. For $c = 2$ these are the classical cubic partition numbers $1, 1, 3, 4, 9, 12, 23, 31, 54, 73, 118, 159, \dots$ (OEIS A002513 [11]).

Remark. The prose sentence accompanying (1) in [2] says that each part may appear in c colours, i.e. describes $1/f_1^c$, and drops the word “even”. That reading is not the one the authors computed with: under $1/f_1^c$ the series at $c = 2$ is not A002513, and *none* of the nine cells below vanishes even at $n = 0$. The displayed form (1) is confirmed by the modular-forms table of [2], which carries $\eta(2z)^{c-1}$ in every row, i.e. the even-part factor f_2^{c-1} , and by Ghoshal and Jana [5], who in restating the definition write that each *even* part may appear in c colours. Everything below uses (1).

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The target is the second of the three conjectures in the concluding remarks of [2], i.e. item 2 of the enumerated list of Section 7, which prints as Conjecture 7.2.

Conjecture 7.2 of [2], as printed in arXiv:2503.19399v2. For all $n \geq 0$, we have

$$\begin{aligned} a_{25}(31^2n + r) &\equiv 0 \pmod{31^2}, & r &\in \{41, 81, 585\}, \\ a_{37}(43^2n + r) &\equiv 0 \pmod{43^2}, & r &\in \{716, 987\}, \\ a_{53}(59^2n + 100) &\equiv 0 \pmod{59^2}, \\ a_{65}(71^2n + r) &\equiv 0 \pmod{71^2}, & r &\in \{41, 47\}, \\ a_{77}(83^2n + 157) &\equiv 0 \pmod{83^2}. \end{aligned}$$

It is a conjunction of nine progressions, one for each triple (c, p, r) listed in Table 2, each with $c = p - 6$; one non-vanishing coefficient falsifies it as printed, and $n = 0$ is admitted by the source’s own quantifier “for all $n \geq 0$ ”. Its lead-in sentence reads “We also have the following conjecture, which cannot be proved similarly like the proof of Theorem [...]”, the cross-reference there being to the proved congruence $a_3(7^2n + 39) \equiv 0 \pmod{7^2}$ of [2].

Locator and scope. The conjecture above is quoted from the arXiv e-print [arXiv:2503.19399v2](#), with only the inter-column spacing reset and the source’s equation label dropped; the nine progressions are otherwise as printed. The block carrying it is byte-identical in v1 and v2. The version of record is behind a paywall with no open-access copy, and its body has not been read: everything below is scoped to the e-print. See §6, item (W1).

2. THE RESULT

Theorem 1. *Every one of the nine progressions of Conjecture 7.2 as printed in arXiv:2503.19399v2 is false. Explicitly:*

- (i) *The fourth line fails at $n = 0$, the smallest index it asserts anything about:*

$$(2) \ a_{65}(41) = 451303321502143296879 \equiv 2170 \pmod{71^2}, \quad 2170 \neq 0,$$

and indeed $a_{65}(41) \equiv 40 \pmod{71}$, so the congruence fails already modulo the first power of the prime.
- (ii) *Each of the other eight lines holds at $n = 0$ and fails at $n = 1$; seven of those eight fail at $n = 1$ already modulo p . The eighteen values are in Table 2.*

Part (i) is the whole refutation of the conjecture as printed, and it is a statement about one coefficient of a power series with 42 terms. Part (ii) is what stops the result being read as a misprint: it is not that eight lines survive and one does not.

The two residues in part (i) are verified by one multiplication each:

$$(3) \quad 451303321502143296879 = 71 \cdot 6356384809889342209 + 40,$$

$$(4) \quad 451303321502143296879 = 5041 \cdot 89526546618159749 + 2170,$$

where $5041 = 71^2$; and $2170 = 71 \cdot 30 + 40$, so the two lines agree.

3. A ONE-INTEGER CERTIFICATE

The following reduction replaces the large integer of (2) by a single small one, and shows that $r = 41$ cannot work for $p = 71$. Here $p(\cdot)$ denotes the ordinary partition function (OEIS A000041 [9]).

Lemma 2. *Let $p \geq 7$ be prime, $c = p - 6$, and put $b(m) := [q^m] f_2^7 / f_1 \in \mathbb{Z}$. Then for every $N \geq 0$*

$$a_c(N) \equiv \sum_{j \geq 0, 2pj \leq N} p(j) b(N - 2pj) \pmod{p}.$$

Proof. Since $(1 - x)^p \equiv 1 - x^p \pmod{p}$ we have $f_2^p \equiv f_{2p} \pmod{p}$, and f_2^7 has constant term 1, so $f_2^{-7} \in \mathbb{Z}[[q]]$ and $f_2^{c-1} = f_2^{p-7} = f_2^p / f_2^7 \equiv f_{2p} / f_2^7 \pmod{p}$. Hence

$$\frac{1}{f_1 f_2^{c-1}} \equiv \frac{f_2^7}{f_1} \cdot \frac{1}{f_{2p}} \pmod{p}.$$

Now $1/f_{2p} = \sum_{j \geq 0} p(j) q^{2pj}$; comparing coefficients of q^N gives the claim. $\square$

The displayed congruence in that proof is Equation (14) of [6], so the reduction is not new; it is restated here only to fix the notation used below.

Corollary 3. *If $N < 2p$ then $a_{p-6}(N) \equiv b(N) \pmod{p}$.*

Apply Corollary 3 with $p = 71$, $c = 65$ (so $c - 1 = 64 = 71 - 7$) and $N = 41 < 142$. Since f_2^7 carries only even exponents, writing $e(k) := [x^k] \prod_{n \geq 1} (1 - x^n)^7$, so that $[q^{2k}] f_2^7 = e(k)$, the sum has exactly 21 terms and nothing is omitted:

$$(5) \quad b(41) = \sum_{k=0}^{20} e(k) p(41 - 2k).$$

Both rows are published sequences: e is OEIS A000730 [10] and p is A000041. Table 1 prints them and the products.

Therefore $b(41) = -31$ and $a_{65}(41) \equiv -31 \equiv 40 \pmod{71}$. Since $71 \nmid 31$, $a_{65}(41)$ is not divisible by 71, a fortiori not by 71^2 , and the conjecture as printed is false.

k	0	1	2	3	4	5	6	7	8	9	10
$e(k)$	1	-7	14	7	-49	21	35	41	-49	-133	98
$p(41-2k)$	44583	31185	21637	14883	10143	6842	4565	3010	1958	1255	792
k	11	12	13	14	15	16	17	18	19	20	
$e(k)$	-21	126	112	-176	-105	-126	140	-35	147	259	
$p(41-2k)$	490	297	176	101	56	30	15	7	3	1	

TABLE 1. The certificate (5). The 21 products are +44583, -218295, +302918, +104181, -497007, +143682, +159775, +123410, -95942, -166915, +77616, -10290, +37422, +19712, -17776, -5880, -3780, +2100, -245, +441, +259; the positive ones sum to 1016099 and the negative ones to 1016130, so $b(41) = -31$. Spot checks from $\prod(1-x^n)^7$: $e(0) = 1$, $e(1) = -7$, $e(2) = 21 - 7 = 14$, $e(3) = -35 + 49 - 7 = 7$.

c	p	r	$n = 0$	$n = 1$	$n = 2$	$n = 1 \bmod p$
25	31	41	0	87	154	25
25	31	81	0	425	685	22
25	31	585	0	336	823	26
37	43	716	0	382	199	38
37	43	987	0	301	288	0
53	59	100	0	1768	3210	57
65	71	41	2170	3779	1478	16
65	71	47	0	3518	885	39
77	83	157	0	429	6028	14

TABLE 2. $a_c(p^2n + r) \bmod p^2$ for the nine printed cells, and the $n = 1$ column reduced mod p . Every line is false. Eight of the nine $n = 1$ values are not even divisible by p ; the single exception is $(c, p, r) = (37, 43, 987)$, where $301 = 43 \cdot 7$ is divisible by 43 and not by 43^2 , so that line is false all the same.

4. ALL NINE LINES

Table 2 evaluates all nine printed cells at $n = 0, 1, 2$.

All 18 221 residue classes $r \in [0, p^2)$ for these five primes, with $c = p - 6$, were tested at $n = 0, 1, 2$, with the following outcome.

Corollary 4. *For each of the five primes $p \in \{31, 43, 59, 71, 83\}$ and $c = p - 6$ there is no residue r modulo p^2 whatsoever for which $a_c(p^2n + r) \equiv 0 \pmod{p^2}$ holds for the three consecutive values $n = 0, 1, 2$. In particular no re-choice of any of the nine printed residues rescues any of the five lines.*

5. RELATED PUBLISHED RESULTS

Normalising the relevant eta-quotient, $\eta(2z)^7/\eta(z) = q^{13/24}f_2^7/f_1$, so by Lemma 2 the governing variable is $24N + 13$. Guadalupe's published mod- p

congruence for this family concerns the residues with

$$(6) \quad 24r + 13 \equiv 0 \pmod{p},$$

and *criterion (6) is not ours*. It is equivalent to $r \equiv 13(p^2 - 1)/24 \pmod{p}$, and in that form it is Guadalupe's Theorem 3.2 of [6] (Theorem 3 of the published version): for every prime $p \geq 7$ with $p \equiv 3, 7 \pmod{8}$, $a_{p-6}(pn + 13(p^2 - 1)/24) \equiv 0 \pmod{p}$ for all $n \geq 0$; which rests in turn on Ahlgren's identity [1, p. 223] for the same series f_2^7/f_1 . That theorem is a *sufficient* condition for mod- p divisibility, and we neither claim nor use any converse: (6) is used here only as a diagnostic, and the nine cells are settled directly by Theorem 1 and Table 2, not by it.

None of Conjecture 7.2's nine residues satisfies (6): in printed order, $(24r + 13) \bmod p$ equals 5, 4, 10, 40, 8, 53, 3, 5, 46. So none of the nine is an instance of Guadalupe's theorem: at $p = 71$ its class is $13(71^2 - 1)/24 = 2730 \equiv 32 \pmod{71}$, i.e. the already-proved line $a_{65}(71n + 32)$ of [2], *not* the witness class 41.

Adjacent work, and what it does not settle. [6] is the closest item: besides the criterion above, its Equation (14) is the first step of Lemma 2, as noted there. It proves a mod- p family in the class just described, and treats neither the modulus p^2 nor any of the nine residues of Conjecture 7.2. Amdeberhan, Sellers and Singh [3] introduce a_c and prove congruences modulo a prime, not modulo p^2 . Ghoshal and Jana [5] confirm Conjecture 7.3 of [2], the overcubic lines modulo 6 and 12: a different numbered statement. Dockery [4] treats powers of 5, Guadalupe [7] the modulus 5, and Thejitha, Sellers and Fathima [8] a two-parameter generalisation $a_{r,s}$; none of the three addresses prime-square moduli or the primes named here. We have found no work recording the failure of any of the nine progressions, and claim no priority for it: what is offered here is a finite computation, and the tools it uses are the cited ones.

6. WHAT IS NOT SETTLED

- (W1) *The body of the version of record has not been read.* The refuted text is the arXiv e-print. The journal version is behind a paywall with no open-access full text available, so nothing here certifies that the published revision still prints these nine residues. Two facts bound that risk without removing it: the conjecture block is byte-identical in v1 and v2, and by Corollary 4 every residue-level variant of the statement is false as well.
- (W2) *The residue sweep is complete in the class but only three terms deep.* All 18 221 classes were tested, but at $n = 0, 1, 2$ only, so the honest form of that negative is exactly Corollary 4. Theorem 1(i) and §3 carry no depth dependence whatever.
- (W3) *The sweep is confined to these five primes with $c = p - 6$,* and says nothing about other pairs (c, p) ; a true congruence of the same shape does exist elsewhere, namely the proved theorem $a_3(7^2n + 39) \equiv 0$

(mod 7^2) of [2], which the same code rediscovers as the unique survivor over all 49 classes to depth 52. Theorem 1 is a finite verification of explicitly exhibited coefficients, not a structural result, and nothing here bears on the proved theorems of [2].

7. VERIFICATION

The accompanying program `verify.py`, with its recorded run in `verify.output.txt`, re-derives from the definition (1) every quantity this note asserts: the exact integer $a_{65}(41)$ of (2), its residues and the division identities (3)–(4); the A000730 row and the dot product (5) giving $b(41) = -31$; Lemma 2, numerically at $p = 71$ for $N = 0, \dots, 1050$ with zero mismatches in a range where 978 of the 1051 values are non-zero mod 71; all 27 entries of Table 2 and its mod- p column; the sweep over all 18 221 classes behind Corollary 4; and the arithmetic of (6). It also checks the coefficient routine against published data (A000041, A002513) and against the source’s own proved congruences $a_{p-6}(pn + r_p) \equiv 0 \pmod{p}$ at $p = 43, 59, 71, 83$, with no exception. It uses the Python standard library only, exact integer arithmetic throughout, no floating point and no randomness, and reports

VERDICT: ALL 55 CHECKS PASS

Its closing scope statement, printed in `verify.output.txt`, lists what it does not cover: nothing bibliographic is re-derived, the paywalled version of record is not consulted, and the residue sweep is not extended past depth 3 or past the five primes named.

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